Class 150, Cast, Corrosion–Resistant, Flanged–End Gate Valves

API STANDARD 603
FIFTH EDITION, JULY 1991

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005

COPYRIGHT 2000 American Petroleum Institute
June 04, 2000 23:17:23
Information Handling Services, 200

Class 150, Cast, Corrosion–Resistant, Flanged–End Gate Valves

Refining Department

API STANDARD 603
FIFTH EDITION, JULY 1991

American
Petroleum
Institute

COPYRIGHT 2000 American Petroleum Institute
June 04, 2000 23:17:23
Information Handling Services, 200

SPECIAL NOTES

1. API PUBLICATIONS NECESSARILY ADDRESS PROBLEMS OF A GENERAL NATURE. WITH RESPECT TO PARTICULAR CIRCUMSTANCES, LOCAL, STATE, AND FEDERAL LAWS AND REGULATIONS SHOULD BE REVIEWED.

2. API IS NOT UNDERTAKING TO MEET THE DUTIES OF EMPLOYERS, MANUFACTURERS, OR SUPPLIERS TO WARN AND PROPERLY TRAIN AND EQUIP THEIR EMPLOYEES, AND OTHERS EXPOSED, CONCERNING HEALTH AND SAFETY RISKS AND PRECAUTIONS, NOR UNDERTAKING THEIR OBLIGATIONS UNDER LOCAL, STATE, OR FEDERAL LAWS.

3. INFORMATION CONCERNING SAFETY AND HEALTH RISKS AND PROPER PRECAUTIONS WITH RESPECT TO PARTICULAR MATERIALS AND CONDITIONS SHOULD BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

4. NOTHING CONTAINED IN ANY API PUBLICATION IS TO BE CONSTRUED AS GRANTING ANY RIGHT, BY IMPLICATION OR OTHERWISE, FOR THE MANUFACTURE, SALE, OR USE OF ANY METHOD, APPARATUS, OR PRODUCT COVERED BY LETTERS PATENT. NEITHER SHOULD ANYTHING CONTAINED IN THE PUBLICATION BE CONSTRUED AS INSURING ANYONE AGAINST LIABILITY FOR INFRINGEMENT OF LETTERS PATENT.

5. GENERALLY, API STANDARDS ARE REVIEWED AND REVISED, REAFFIRMED, OR WITHDRAWN AT LEAST EVERY FIVE YEARS. SOMETIMES A ONE-TIME EXTENSION OF UP TO TWO YEARS WILL BE ADDED TO THIS REVIEW CYCLE. THIS PUBLICATION WILL NO LONGER BE IN EFFECT FIVE YEARS AFTER ITS PUBLICATION DATE AS AN OPERATIVE API STANDARD OR, WHERE AN EXTENSION HAS BEEN GRANTED, UPON REPUBLICATION. STATUS OF THE PUBLICATION CAN BE ASCERTAINED FROM THE API AUTHORING DEPARTMENT [TELEPHONE (202) 682-8000]. A CATALOG OF API PUBLICATIONS AND MATERIALS IS PUBLISHED ANNUALLY AND UPDATED QUARTERLY BY API, 1220 L STREET, N.W., WASHINGTON, D.C. 20005.

Copyright © 1991 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

FOREWORD

This standard is a purchase specification of requirements for high-alloy, corrosion-resistant, flanged-end gate valves for use in petroleum refinery piping systems.

A valve purchased in accordance with this standard may have a wall thickness less than that required by API Standard 600. While highly corrosion resistant, a valve purchased in accordance with this standard may provide less corrosion allowance than a valve manufactured in accordance with API Standard 600. If more corrosion allowance is required, it is recommended that the valve conform to the wall thickness requirements of API Standard 600.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or incorporating sections thereof into another complete standard.

API standards are published as an aid to procurement of standardized equipment and materials. These standards are not intended to inhibit purchasers or producers from purchasing or producing products made to specifications other than those of API.

API publications may be used by anyone desiring to do so. Every effort has been made by the Institute to assure the accuracy and reliability of the data contained in them; however, the Institute makes no representation, warranty, or guarantee in connection with this publication and hereby expressly disclaims any liability or responsibility for loss or damage resulting from its use or for the violation of any federal, state, or municipal regulation with which this publication may conflict.

Suggested revisions are invited and should be submitted to the director of the Refining Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

IMPORTANT INFORMATION CONCERNING USE OF ASBESTOS OR ALTERNATIVE MATERIALS

Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

SAFETY AND HEALTH INFORMATION WITH RESPECT TO PARTICULAR PRODUCTS OR MATERIALS CAN BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT PRODUCT OR MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

CONTENTS

		Page
Notes to Purchaser		vii
SECTION 1—GENERAL		
1.1	Scope	1
1.2	Referenced Publications	1
1.3	Pressure–Temperature Ratings	1
SECTION 2—DESIGN		
2.1	Body	1
2.2	Bonnet	3
2.3	Gate	3
2.4	Yoke	4
2.5	Handwheel and Handwheel Nut	4
2.6	Stem and Stem Nut	4
2.7	Stuffing Box, Packing, and Lantern Ring	5
2.8	Bolting	5
2.9	Operation	6
2.10	Bypass	6
2.11	Safety Shield	6
SECTION 3—MATERIAL		
3.1	Shell	6
3.2	Body Seat Rings	6
3.3	Bonnet Gasket	6
3.4	Gate	6
3.5	Yoke	6
3.6	Handwheel, Chainwheel, and Handwheel Nut	6
3.7	Stem Nut	7
3.8	Gland	7
3.9	Trim	7
3.10	Stem Packing	7
3.11	Bolting	7
3.12	Bypass	7
3.13	Safety Shield	7
3.14	Nameplate	7
SECTION 4—EXAMINATION, INSPECTION, AND TESTING		
4.1	Inspection	7
4.2	Pressure Tests	7
4.3	Repair of Defects	7
SECTION 5—SHIPMENT		
5.1	Marking	8
5.2	Painting	8
5.3	Openings	8
5.4	Gate Position and Lubricant	8
5.5	Stem Packing	8
5.6	Packaging	8

Figures

1	Typical Gate Valve	2
2	Types of Gates	4

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

Page

3—Wear Travel of a Wedge Gate 4
4—Alternative Designs of the Safety Shield 6

Tables
1—Valve Dimensions 2
2—Minimum Thread Length of Tapped Test Opening 3
3—Minimum Size of Bonnet-Flange Through Bolt 3
4—Wear Travel 4
5—Undertolerance for Diameter of Stem in Packing Area and for Major Diameter of Operating Threads 5
6—Stuffing Box and Packing Dimensions 5

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

NOTES TO PURCHASER

1. If the purchaser needs a corrosion-resistant valve that deviates from this standard, the deviating requirements shall be stated in the purchase order.
2. If no exceptions are to be made to this standard, the purchase order need only refer to API Standard 603 and specify the items marked with an asterisk (*) in the following list. Other items listed are options that may also be specified.

* a. Valve size (see 1.1.1).
  b. Hard-faced body seats, if required (see 2.1.3).
  c. Flat-face end flanges, if required (see 2.1.5).
  d. Size and location of any drain connections required (see 2.1.6).
  e. Hard-faced stem-hole guide, if required (see 2.2.4).
* f. Whether the gate is a wedge gate or double-disc gate (see 2.3.1).
  g. If a one-piece wedge gate is required, whether it is a solid-wedge gate or a flexible-wedge gate (see 2.3.2 and 2.3.3).
  h. Hard-faced gate-seating surfaces, if required (see 2.3.7 and 3.9.3).
  i. Lantern ring, if required (see 2.7.4).
  j. If the valve is not operated directly by a handwheel, how it is operated (see 2.9.1).
  k. If chainwheel operation is required, the type of chainwheel and chain (see 2.9.2).
  l. If gear operation is required, the type and arrangement of gears and the design maximum differential pressure across the valve (see 2.9.3).
  m. If power operation is required, the type of power, the type of power unit, and the design maximum differential pressure across the valve (see 2.9.4).
  n. Bypass, if required (see 2.10.1).
  o. Safety shield, if required (see 2.11).
* p. Valve shell material (see 3.1).
  q. For valves in corrosion-resistant service, the bonnet gasket material (see 3.3).
  r. For a yoke separate from the bonnet, whether a specific material is required (see 3.5).
  s. For valves in corrosion-resistant service, the stem packing material (see 3.10).
  t. If safety shield material other than 18Cr-8Ni steel is required, the specific material required (see 3.13).
  u. Inspection by the purchaser, if required (see note 5 below and 4.1).
  v. Special painting, if required (see 5.2).
  w. Export packaging, if required (see 5.6).
  x. Tapped test openings, if required (see 2.1.7).

3. If a valve is to be used at design temperatures below –20°F (–29°C), the purchaser shall specify any special requirements for impact testing of the shell and stem material; for bolting of the bonnet, yoke, and gland; for packing; and for the bonnet gasket.
4. As indicated in the foreword, the valve described in this standard may have a wall thickness less than that of the valves described in API Standard 600.
5. Refer to API Standard 598 for additional items that may need to be specified, including the extent of inspection, the inspector's address, and optional high-pressure closure tests.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

Class 150, Cast, Corrosion-Resistant, Flanged-End Gate Valves

SECTION 1—GENERAL

1.1 Scope

1.1.1 This standard covers Class 150, cast, corrosion-resistant, flanged-end gate valves in sizes NPS ½ through 12. These sizes correspond to nominal pipe sizes in ASME B36.10M.

1.1.2 The valves are outside-screw-and-yoke (OS&Y) valves with rising stem, nonrising handwheel, bolted bonnet, integral seats in the body and on the gate or separate seats in the body, raised-face end flanges, and various types of gates.

1.1.3 Figure 1 illustrates a typical acceptable design for a gate valve and shows the standard nomenclature for valve parts.

1.1.4 The dimensions in customary units are standard. Metric (SI) units are shown for reference.

1.2 Referenced Publications

The most recent edition or revision of the following standards or specifications shall, to the extent specified in this standard, form a part of this standard:

ASME[1]

B1.1 *Unified Inch Screw Threads (UN and UNR Thread Form)*
B1.5 *Acme Screw Threads*
B1.8 *Stub Acme Screw Threads*
B1.12M *Class 5 Interference-Fit Thread*
B16.5 *Pipe Flanges and Flanged Fittings*
B16.10 *Face-to-Face and End-to-End Dimensions of Valves*
B16.11 *Forged Steel Fittings, Socket-Welding and Threaded*
B16.34 *Valves—Flanged, Threaded, and Welding End*
B18.2.2 *Square and Hex Nuts (Inch Series)*
B36.10M *Welded and Seamless Wrought Steel Pipe*

API

Std 598 *Valve Inspection and Testing*
Std 600 *Steel Gate Valves—Flanged and Butt-Welding Ends*

MSS[2]

SP-42 *Class 150 Corrosion Resistant Gate, Globe, Angle and Check Valves with Flanged and Butt Weld Ends*

ASTM[3]

A 193 *Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service*
A 194 *Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service*

AWS[4]

A5.13 *Solid Surfacing Welding Rods and Electrodes*

1.3 Pressure–Temperature Ratings

1.3.1 Pressure–temperature ratings for Class 150 valves covered by this standard shall be those listed for the appropriate material under Standard Class in ASME B16.34. Each pressure rating is the maximum allowable sustained non-shock pressure at the corresponding tabulated temperature. Linear interpolation is permitted.

1.3.2 The design metal temperature for determining a pressure rating (the maximum allowable pressure at a design temperature) shall be selected by the purchaser and is subject to the requirements of any applicable code.

SECTION 2—DESIGN

2.1 Body

2.1.1 The wall thickness of the body shall not be less than that specified in ASME B16.34 for Class 150 valves (shown in Table 1).

2.1.2 The face-to-face dimension shall be that specified in ASME B16.10 for a Class 150 raised-face gate valve (shown in Table 1).

2.1.3 Except as permitted below, the body seat shall be integral with the body. Body seats shall not be hard faced unless specified in the purchase order. If hard facing is specified, it shall be deposit welded to provide a minimum finished thickness of 0.06 inch (1.6 millimeter). For NPS 4 and smaller valves in which it is impractical to weld hard-faced integral body seats, hard facing may be deposit

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

[2]Manufacturers Standardization Society of the Valve and Fittings Industry, 127 Park Street, N.E., Vienna, Virginia 22180.

[3]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

[4]American Welding Society, 550 N.W. LeJeune Road, Miami, Florida 33135.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

API STD*603 91 ■ 0732290 0099703 6 ■

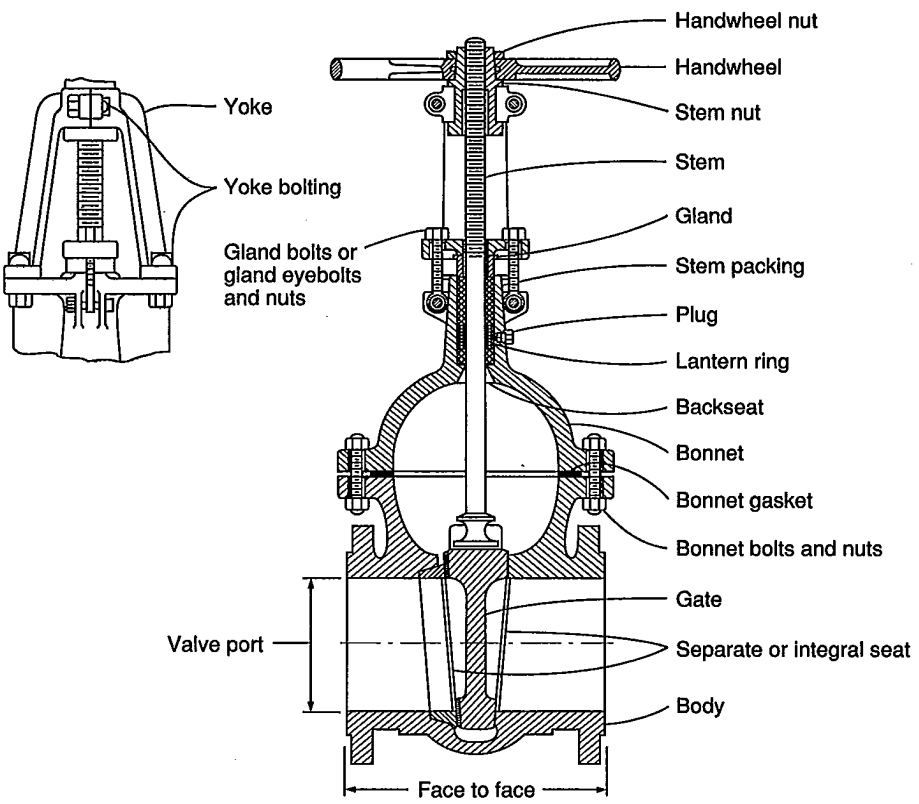


Figure 1—Typical Gate Valve

welded onto separate seat rings threaded, pressed, or welded into the body.

2.1.4 The nominal bore diameter in inches, including the diameter of end ports and body seats, shall equal the nominal valve size.

2.1.5 End flanges shall be cast integrally with the body. They shall be raised face unless otherwise specified in the purchase order. For NPS 1 and larger valves, the flange thickness, outside diameter, drilling, tolerances, facing finish, and spot or back facing shall be as specified in ASME

Table 1—Valve Dimensions

Valve Size (NPS)	Minimum Shell[a] Wall Thickness		Face-to-Face of End Flanges		Minimum Stem Outside Diameter[b]		Minimum Handwheel Outside Diameter	
	Inches	Millimeters	Inches	Millimeters	Inches	Millimeters	Inches	Millimeters
½	0.11	2.8	4¼	108	7/16	11.1	3	76
¾	0.12	3.0	4⅝	118	7/16	11.1	3½	89
1	0.16	4.0	5	127	½	12.7	4	102
1¼	0.19	4.8	5½	140	½	12.7	4	102
1½	0.19	4.8	6½	165	9/16	14.3	4¾	121
2	0.22	5.6	7	178	5/8	15.9	5	127
2½	0.22	5.6	7½	191	5/8	15.9	7	178
3	0.22	5.6	8	203	¾	19.0	7½	191
4	0.25	6.4	9	229	⅞	22.2	9	229
6	0.28	7.1	10½	267	1	25.4	10	254
8	0.31	7.9	11½	292	1⅛	28.6	12	305
10	0.34	8.7	13	330	1¼	31.8	14	356
12	0.38	9.5	14	356	1⅜	35.0	16	406

[a]The shell comprises the body and the bonnet.

[b]The diameters listed are the diameter of the stem in the packing area and the major diameter of the operating threads. An undertolerance is permitted as shown in Table 5.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

B16.5 for Class 150 flanged fittings. For NPS ½ and ¾ valves, flanges shall be as specified in MSS SP-42.

2.1.6 Drain connections in the body shall be made only if specified in the purchase order. If required for a valve NPS 2 or larger, drain connections shall be sized and located per ASME B16.34 unless otherwise specified in the purchase order.

2.1.7 Tapped test openings are permitted only if specified in the purchase order and shall not be larger than NPS ½. For NPS ½ through 1 valves, the tap may be NPS ⅛. For other valve sizes, the tap shall not be smaller than NPS ¼. After testing, the tapped hole shall be fitted with a threaded, solid plug as specified in ASME B16.11 and of the same nominal composition as the shell. The test tap may require a boss to provide the minimum thread length indicated in Table 2.

2.1.8 Body design shall ensure that the gate is guided to minimize wear during valve operation (see 2.3.6).

2.2 Bonnet

2.2.1 The minimum wall thickness of the bonnet, except for the bonnet neck above the backseat, shall be as specified in Table 1 for the shell. Above the backseat, the bonnet neck shall conform to the design requirements of ASME B16.34.

2.2.2 The bonnet shall have a bolted, flanged joint with at least four through bolts of the size listed in Table 3. The total cross-sectional area of the bolts shall be as required by ASME B16.34. The flanges shall be flat face, raised face, or tongue and groove. The facing finish shall be as specified for flanges in ASME B16.5. The gasket shall not extend beyond the inner edge of the bolt holes.

2.2.3 A machined conical or spherical backseat shall be provided in the bonnet to contact a corresponding seating surface on the valve stem. The backseat may have either an integral surface or weld-deposited hard facing with a minimum finished thickness of 0.06 inch (1.6 millimeter).

2.2.4 The stem hole in the bonnet shall be designed to provide proper clearance for guiding the stem and to prevent packing extrusion. The stem-hole guide shall not be hard faced unless specified in the purchase order. When hard facing is specified, it shall have a minimum finished thickness of 0.06 inch (1.6 millimeter). (See 3.9.3.)

Table 2—Minimum Thread Length of Tapped Test Opening

Test Opening Size (NPS)	Minimum Thread Length		Boss Diameter (When Required)	
	Inches	Millimeters	Inches	Millimeters
⅛	⅜	9.5	1	25.4
¼	⅜	9.5	1	25.4
⅜	7/16	11.1	1¼	31.8
½	½	12.7	1½	38.1

Table 3—Minimum Size of Bonnet-Flange Through Bolt

Valve Size (NPS)	Minimum Bolt Size (Nominal)	
	Inches	Millimeters
½ to 2½	⅜	9.5
3 to 8	½	12.7
10 to 12	⅝	15.9

2.2.5 If a tapped opening is made in the bonnet to test the valve, it shall conform to the requirements of 2.1.7.

2.3 Gate

2.3.1 The types of gates are illustrated in Figure 2 and are classified as follows:

a. Wedge gate, including one-piece solid wedge, one-piece flexible wedge, and two-piece split wedge.
b. Double-disc gate.

2.3.2 If a wedge gate is specified in the purchase order, a split-wedge gate shall be furnished unless a one-piece wedge gate is specified. Wedges shall have a nominal taper of 10 degrees (included angle), and the fully opened gate shall completely clear the valve ports.

2.3.3 A one-piece wedge gate may be made by welding. It may be either a solid-wedge gate with a trapezoidal or I-shaped cross section or a flexible-wedge gate with a tapered cross section. If the purchase order does not specify the particular type of one-piece wedge gate, either type may be furnished.

2.3.4 A split-wedge gate valve is a valve with a two-piece gate whose pieces are positioned according to the manufacturer's choice. The valve design shall ensure that the pieces cannot separate regardless of the gate position or valve orientation.

2.3.5 A double-disc gate valve shall have parallel seats and an internal spreading device that will force the discs firmly against the body seats when the gate reaches the closed position.

2.3.6 The gate shall have guide surfaces to minimize seat wear during valve operation and to properly guide the gate when the valve is installed in any position.

2.3.7 Gate-seating surfaces shall be integral with the gate. They shall not be hard faced unless specified in the purchase order. When hard facing is specified, it shall have a minimum finished thickness of 0.06 inch (1.6 millimeter).

2.3.8 Gate-seating design shall ensure adequate seating width of the seating surfaces both before and after wear. This shall be ensured by proportioning the width of the body and gate seats and by seating the new gate high on the body seats

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

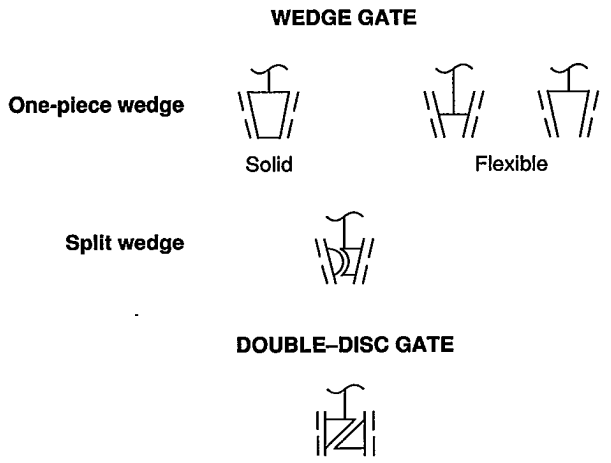


Figure 2—Types of Gates

so that the gate seats will still completely contact the full width of the body seats after the gate in the closed position has traveled down (because of seat wear) by a distance (wear travel) not less than that listed in Table 4. Wear travel of a wedge gate is illustrated in Figure 3.

2.4 Yoke

The yoke may be either integral with or separate from the bonnet. If the yoke is separate, the yoke-to-bonnet joint shall be machined, and the yoke shall be bolted to the bonnet with through bolts.

2.5 Handwheel and Handwheel Nut

2.5.1 For a valve that is operated directly by a handwheel and that is NPS 1 or smaller, the handwheel shall be spoked or with a ventilated or knobbed rim. For a valve that is operated directly by a handwheel and that is NPS 1½ or larger, the handwheel shall be spoked.

2.5.2 The minimum size of the handwheel (taken as the outside diameter of the rim) shall be as specified in Table 1.

Table 4—Wear Travel

Valve Size (NPS)	Minimum Wear Travel	
	Inches	Millimeters
½ to 2	0.09	2.3
2½ to 6	0.13	3.3
8 to 12	0.25	6.4

2.5.3 The handwheel rim shall have the word *open* and an arrow pointing in the direction to open the valve unless rim size or design make them impracticable.

2.5.4 The handwheel shall be fixed to the stem nut by a threaded, hexagonal or octagonal, handwheel nut.

2.6 Stem and Stem Nut

2.6.1 The minimum outside diameter of the stem is shown in Table 1. To allow the use of standard diameter round bars, an undertolerance is permitted in accordance with Table 5.

2.6.2 The stem connection to the gate shall be flexible and of a T-head or threaded-and-pinned type. The stem connection shall prevent turning of the stem in the gate or disengagement of the stem from the gate while in service. The strength (in tension) under axial load of the stem-to-gate connection and of all parts of the stem within the pressure boundary shall exceed the strength of the stem at the root of its operating thread.

2.6.3 The stem thread shall be a single-lead or double-lead thread and shall be of the left-hand Acme type as specified in ASME B1.5 or ASME B1.8; minor modifications are permitted.

2.6.4 The stem shall be raised by a rotating stem nut mounted at the top of the yoke. The stem nut shall have a hexagonal or round shank with a keyway for attachment of the handwheel.

2.6.5 The stem shall project far enough outside the stem nut on a new closed valve that after seat wear has allowed the gate to travel to its lowest closed position, the stem thread will fully engage the stem nut thread.

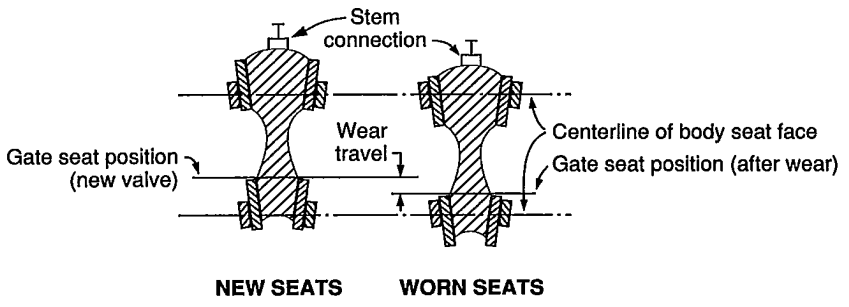


Figure 3—Wear Travel of a Wedge Gate

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

Table 5—Undertolerance for Diameter of Stem in Packing Area and for Major Diameter of Operating Threads

Diameter		Undertolerance	
Inches	Millimeters	Inches	Millimeters
To 5/8	To 15.9	0.012	0.31
Over 5/8 to 7/8	Over 15.9 to 22.2	0.013	0.33
Over 7/8 to 1	Over 22.2 to 25.4	0.014	0.36
Over 1 to 1 1/8	Over 25.4 to 28.6	0.015	0.38
Over 1 1/8 to 1 1/4	Over 28.6 to 31.8	0.016	0.41
Over 1 1/4 to 1 3/4	Over 31.8 to 34.9	0.017	0.43
Over 1 3/4 to 1 1/2	Over 34.9 to 38.1	0.019	0.48

2.6.6 The stem shall have an integral annular backseat surface to seat against the bonnet backseat when the gate is fully open. Valves furnished with the stem attached to the gate by a gate-spreading device may have the backseat machined on the top surface of the device if the connection with the stem is designed to prevent leakage past the stem when the valve is backseated.

2.7 Stuffing Box, Packing, and Lantern Ring

2.7.1 The dimensions of the stuffing box shall be as specified in Table 6.

2.7.2 The packing gland shall be flanged with holes (not slots) for two gland bolts. The gland shall be a solid one-piece design or a two-piece self-aligning design consisting of a gland proper and a separate gland flange.

2.7.3 The nominal packing size shall be as specified in Table 6 (see 5.5).

2.7.4 The stuffing box shall not be furnished with a lantern ring unless specified in the purchase order. If a lantern ring is furnished, its length shall not be less than the nominal diameter of the stem. For square packing of the width shown in Table 6, the minimum depth of the stuffing box shall be increased by the sum of the length of the lantern ring and the depth required for the packing below the lantern ring. If a lantern ring is furnished, the stuffing box shall be tapped opposite the center of the installed lantern ring, fitted with a threaded pipe plug at least NPS 1/4 as specified in ASME B16.11, and provided with a boss, as required, similar to that specified in 2.1.7 for a test tap in the body.

2.8 Bolting

2.8.1 Bonnet and yoke bolting shall be headed bolts or continuously threaded stud bolts with hexagonal nuts conforming to ASME B18.2.2.

2.8.2 Gland bolting shall be headed bolts, hinged eyebolts, stud bolts, or studs with hexagonal nuts.

2.8.3 Bolting 1 inch in diameter or smaller shall have coarse threads. Bolting larger than 1 inch shall be 8-thread series (8 UN). Bolt threads shall be Class 2A and nut threads

Table 6—Stuffing Box and Packing Dimensions

Stem Outside Diameter[a]		Nominal Packing Width		Stuffing Box Dimensions			
				Nominal Bore[b]		Minimum Depth[c]	
Inches	Millimeters	Inches	Millimeters	Inches	Millimeters	Inches	Millimeters
7/16	11.1	5/32	4.0	3/4	19.0	25/32	19
1/2	12.7	3/16	4.8	7/8	22.2	15/16	23
9/16	14.3	3/16	4.8	15/16	23.8	15/16	23
5/8	15.9	1/4	6.4	1 1/8	28.6	1 1/4	31
11/16	17.5	1/4	6.4	1 3/16	30.2	1 1/4	31
3/4	19.1	1/4	6.4	1 1/4	31.8	1 1/4	31
7/8	22.2	1/4	6.4	1 3/8	35.0	1 1/4	31
1	25.4	1/4	6.4	1 1/2	38.1	1 1/4	31
1 1/8	28.6	5/16	7.9	1 3/4	44.4	1 9/16	39
1 1/4	31.8	5/16	7.9	1 7/8	47.6	1 9/16	39
1 3/8	34.9	5/16	7.9	2	50.8	1 9/16	39
1 1/2	38.1	3/8	9.5	2 1/4	57.2	1 7/8	47
1 5/8	41.3	3/8	9.5	2 3/8	60.3	1 7/8	47

[a]For stem outside diameters not listed, the packing shall be square and the size shall be in proportion to the sizes shown above.
[b]The tolerance for the bore diameter is ±1/64 inch (±0.40 millimeters).
[c]This depth is equivalent to that of five rings of uncompressed square packing, the depth required if no lantern ring is furnished.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

Class 2B in accordance with ASME B1.1; but studs used for gland bolting shall be fitted into the stuffing box flange with a Class 5 interference-fit in accordance with ASME B1.12M.

2.9 Operation

2.9.1 Unless otherwise specified in the purchase order, the valve shall be operated directly by a handwheel.

2.9.2 If chainwheel operation is required, the type of chainwheel and chain required shall be specified in the purchase order.

2.9.3 If gear operation is required, the type of gears, their arrangement, and the design maximum differential pressure across the valve shall be specified in the purchase order.

2.9.4 If power operation is required, the type of power, the power unit, and the design maximum differential pressure across the valve shall be specified in the purchase order.

2.10 Bypass

2.10.1 A bypass shall not be furnished unless specified in the purchase order.

2.10.2 Unless otherwise specified in the purchase order, any bypass that is furnished shall be of the valve external type and of a size that is specified in ASME B16.34. The bypass shall be located on the side of the valve connecting either the A–B or the E–F locations shown in ASME B16.34. The bypass valve stem shall have the same general orientation as the primary valve stem.

2.10.3 Unless otherwise specified in the purchase order, the bypass valve shall be an outside-screw-and-yoke globe valve with a flanged or union bonnet.

2.11 Safety Shield

2.11.1 When specified in the purchase order, a safety shield to protect the operator shall be furnished.

2.11.2 When a safety shield is furnished, it shall be mounted on the stem nut under the handwheel or between the top of the yoke and the stem-nut retaining nut, or cap. The safety shield shall have an outside diameter not less than that of the handwheel. The shield shall have one of the two alternative shapes shown in Figure 4.

SECTION 3—MATERIAL

3.1 Shell

The body and bonnet shall be cast of the material specified in the purchase order using a material specification listed under Group 2 or 3 materials in ASME B16.34.

3.2 Body Seat Rings

When separate body seat rings are used, they shall be of the same nominal composition as the shell (see 3.9.3).

3.3 Bonnet Gasket

Unless otherwise specified in the purchase order, the bonnet gasket shall be suitable for a valve design temperature of 750°F (399°C).

3.4 Gate

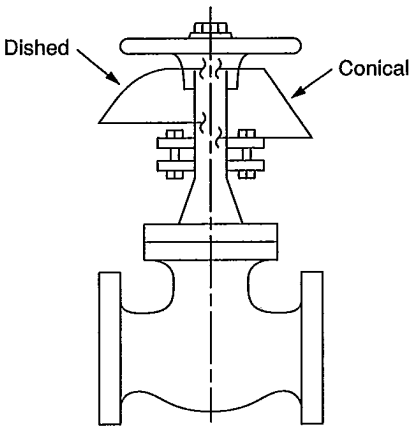
Solid- and flexible-wedge gates and discs for split-wedge or double-disc gates shall be of the same nominal composition as the shell (see 3.9.3).

3.5 Yoke

If the yoke is separate from the bonnet, it shall be made of carbon steel, ductile iron, malleable iron, 18Cr-8Ni steel, or the same material as the shell.

3.6 Handwheel, Chainwheel, and Handwheel Nut

3.6.1 Handwheels and chainwheels shall be made of carbon steel, ductile iron, or malleable iron. Unless otherwise specified in the purchase order, either they shall be cast or forged or they shall be fabricated from other carbon steel product forms that provide strength and toughness



Note: The design of the safety shield shall permit repacking of the valve without removal of the shield.

Figure 4—Alternative Designs of the Safety Shield

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

comparable to those provided by casting or forging. All handwheels shall be free of burrs and sharp edges.

3.6.2 The handwheel nut shall be made of a nonferrous copper alloy, a 13Cr steel, or an 18Cr-8Ni steel.

3.7 Stem Nut

The stem nut shall be made of ductile iron, 13Cr steel, 18Cr-8Ni steel, or copper alloy with a melting point above 1750°F (954°C).

3.8 Gland

The gland (one- or two-piece) shall be of the same nominal composition as the shell.

3.9 Trim

3.9.1 The term *trim* includes the following parts:

a. Stem.
b. Body-seating surface.
c. Gate-seating surface.
d. Hard facing of stem-hole guide.
e. Small internal parts that contact the service fluid.

3.9.2 The stem shall be made of a wrought material with the same nominal composition as the valve body and bonnet material as follows: When the valve shell is made of 18Cr-8Ni-Mo, the stem shall be made of AISI[5] 316 steel. When the valve shell is made of 18Cr-8Ni or 18Cr-8Ni-Cb, the stem shall be made of AISI 304 or 347 steel. When the valve shell is made of other material, the stem shall be made of a material with the same nominal composition as that of the shell.

3.9.3 When specified, the hard facing of gate, body seat, and stem-hole guide shall be in accordance with AWS A5.13, classification CoCr-A, unless other materials are specified in the purchase order.

3.9.4 Small parts of the trim shall be of the same nominal composition as the shell and include the following:

a. Stem connection.
b. Internal pins or screws.
c. Spreading mechanism of a double-disc valve.
d. Plugs for the test or drain taps in the bonnet and body.
e. Lantern ring and plug in stuffing box tap.

3.10 Stem Packing

Unless otherwise specified in the purchase order, the stem packing shall be suitable for a packing design temperature of 750°F (399°C).

3.11 Bolting

3.11.1 Bonnet and yoke bolts shall be 18Cr-8Ni steel bolts, ASTM A 193 Grade B8 bolts with ASTM A 194 Grade 8 nuts, ASTM A 193 Grade B8C bolts with ASTM A 194 Grade 8C nuts, or ASTM A 193 Grade B8M bolts with ASTM A 194 Grade 8M nuts.

3.11.2 Gland bolts and nuts shall be made of 18Cr-8Ni steel.

3.12 Bypass

When specified, the bypass valve and its piping shall be of the same nominal composition as the shell.

3.13 Safety Shield

When specified, the safety shield shall be made of 18Cr-8Ni steel unless otherwise specified.

3.14 Nameplate

The nameplates for valves NPS 6 and larger shall be made of 18Cr-8Ni steel. For smaller valves, nameplates shall be made of the manufacturer's standard metal material, such as brass or aluminum, and shall be resistant to atmospheric corrosion.

SECTION 4—EXAMINATION, INSPECTION, AND TESTING

4.1 Inspection

When inspection by the purchaser is specified in the purchase order, inspection shall be in accordance with API Standard 598. The manufacturer shall examine each valve produced in accordance with API Standard 598.

[5]American Iron and Steel Institute, 1000 16th Street, N.W., Washington, D.C. 20036.

4.2 Pressure Tests

Each valve shall be pressure tested as specified in API Standard 598.

4.3 Repair of Defects

Defects in the shell of a cast valve that are revealed by inspection or testing may be repaired as permitted by the appropriate ASTM cast material specification.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

API STD*603 91 ■ 0732290 0099709 7 ■

SECTION 5—SHIPMENT

5.1 Marking

Valves shall be marked in accordance with the requirements of ASME B16.34.

5.2 Painting

Except for the handwheel or yoke, valve parts shall not be painted. Paint color is optional.

5.3 Openings

5.3.1 End flanges shall be blanked to protect the gasket surfaces and valve internals during shipment and storage. The protective covers shall be made of wood, wood fiber, or plastic and shall be securely attached to the valve ends by bolting, steel straps, steel clips, or suitable friction-locking devices. The design of the protective cover shall ensure that the valve cannot be installed without complete removal of the cover.

5.3.2 Tapped connections shall be fitted with fully tightened threaded plugs (see 2.1.7 and 3.9.4).

5.4 Gate Position and Lubricant

Valves shall be shipped with the gates closed and with the stem threads lubricated.

5.5 Stem Packing

Valves shall be shipped with the packing installed and ready for operation. The remaining adjustment length of the packing gland at the time of shipment, with the gland tight, shall be greater than 1.5 times the packing width specified in Table 6.

5.6 Packaging

5.6.1 If export packaging is not specified in the purchase order, valves may be shipped loose, palletized, or packed in boxes or crates.

5.6.2 If export packaging is specified in the purchase order, valves shall be shipped individually or collectively in wooden boxes or crates in a manner that will prevent shifting within the package.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services, 200
June 04, 2000 23:17:23

Order No. 822-60300

1-1300—7/91—5C (5A)

COPYRIGHT 2000 American Petroleum Institute
June 04, 2000 23:17:23
Information Handling Services, 200

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



COPYRIGHT 2000 American Petroleum Institute
June 04, 2000 23:17:23
Information Handling Services, 200